

Statistical Inference Report

QSR Scheduling System Evaluation

Inferential Statistics and Hypothesis Testing

1 Objective

A restaurant chain is evaluating three scheduling systems:

- Fixed Shift
- Flexible Shift
- AI-Optimized

The goal is to determine whether the scheduling system affects sales performance, employee satisfaction, and turnover.

2 Dataset Description

The dataset contains 90 restaurant locations.

Table 1: Variable Definitions

Variable	Description
location_id	Restaurant identifier
schedule_type	Scheduling system used
weekly_sales_usd	Weekly sales revenue
employee_satisfaction	Satisfaction score (1–5)
turnover_6mo	High turnover (Yes/No)

3 Tasks

1. One-Way ANOVA

Test whether mean weekly sales differ among the three scheduling systems.

Report:

- Group means
- F-statistic
- p-value

2. Kruskal-Wallis Test

Determine whether employee satisfaction differs among the three scheduling systems.

Report:

- H-statistic
- p-value

3. Chi-Square Test

Determine whether turnover rates differ among the three scheduling systems.

Report:

- Chi-square statistic
- p-value

4 Discussion

Briefly answer:

1. Which scheduling system performs best?
2. Does employee satisfaction differ across systems?
3. Does turnover differ across systems?
4. Which system would you recommend and why?

5 Deliverables

1. Analysis script/Notebook (Python).
2. Relevant tables and charts.
3. Short report (2 pages maximum).